In our architecture, the platform team Gray provides is a configurable Terraform template. The team chooses to put up the latest template inside a Git repository. In this case, the Git repository together with its README file becomes the platform interface. The template becomes the platform kernel.

It is not necessary to have a full team working on such a minimal platform. The platform kernel can be upgraded by providing a new version of the Terraform template and letting teams upgrade their stacks. This platform has a very low degree of abstraction right now.

However, the templates already provide a lot of benefits to the data producers as well as consumers, thanks to the architecture of the data products.

8.1.3 Identifying the components of the data product

The data products all share common tooling based on Google Cloud Dataflow and its associated temporary storage. This enables a team to easily create data products. Dataflow allows teams to create pipelines that output data products in Java, Python, or SQL.

For the data product owned by the Black team, the team chooses to directly connect to the sources, using the Dataflow pipeline. Its upstream data sources are Pub/ Sub and the BigQuery dataset.

The Dataflow pipeline produces one BigQuery dataset. This platform uses datasets as containers for data products. One data product has exactly one BigQuery dataset. BigQuery datasets can hold multiple tables and views, but the access permissions can easily be handled at the dataset level. Having multiple tables inside one data product is totally normal. This dataset becomes the first data output port, as you can see in figure 8.2.